

Dielectric Properties of Matter

Polaritons in an ionic crystal

derivation notes

1 Polarisation waves longitudinal vs transverse

For waves travelling in the $+x$ direction:

$$P_x = P_{x0} e^{i(qx - \omega t)} \rightarrow \text{longitudinal wave}$$

$$P_y = P_{y0} e^{i(qx - \omega t)} \rightarrow \text{transverse wave}$$

Longitudinal waves

$$\nabla \times \vec{P}_L = 0, \quad \nabla \cdot \vec{P}_L \neq 0 \quad (1)$$

Transverse waves

$$\nabla \times \vec{P}_T \neq 0, \quad \nabla \cdot \vec{P}_T = 0 \quad (2)$$

The medium responds through

$$\vec{P} = \epsilon_0(\epsilon(\omega) - 1)\vec{E} \quad (3)$$

2 Longitudinal waves result: $\epsilon(\omega_L) = 0$

In a dielectric without any external charge, $\nabla \cdot \vec{D} = 0$. Using $\vec{D} = \epsilon_0 \epsilon \vec{E}$ and $\vec{E} = \vec{P} / [\epsilon_0(\epsilon - 1)]$:

$$\begin{aligned} \nabla \cdot \vec{D} &= 0 \\ \Rightarrow \epsilon_0 \epsilon(\omega) \nabla \cdot \vec{E} &= 0 \\ \Rightarrow \cancel{\epsilon_0} \epsilon(\omega) \frac{\nabla \cdot \vec{P}}{\cancel{\epsilon_0}(\epsilon(\omega) - 1)} &= 0 \\ \Rightarrow \frac{\epsilon(\omega)}{\epsilon(\omega) - 1} \nabla \cdot \vec{P} &= 0 \end{aligned} \quad (4)$$

For longitudinal waves to exist we need $\nabla \cdot \vec{P} \neq 0$, so the prefactor must vanish:

$$\frac{\epsilon(\omega)}{\epsilon(\omega) - 1} = 0 \quad \therefore \quad \boxed{\epsilon(\omega_L) = 0}$$

Again, from $\vec{P} = \epsilon_0(\epsilon(\omega) - 1)\vec{E}$ with $\epsilon(\omega_L) = 0$:

Depolarising field

$$\vec{P}_L = -\epsilon_0 \vec{E}_L \quad \Rightarrow \quad \vec{E}_L = -\frac{\vec{P}_L}{\epsilon_0}$$

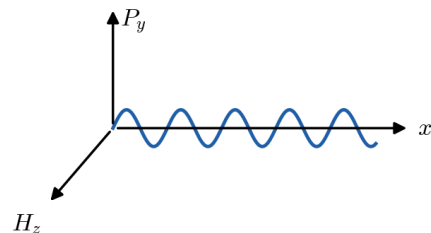
3 Transverse waves solution

Maxwell equations

$$\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad (5)$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \quad (6)$$

with $\vec{E}(\epsilon(\omega) - 1)\epsilon_0 = \vec{P}$



P_y and H_z , wave along x

From (5), replacing $\vec{E}$ by $\vec{P}$:

$$\frac{\nabla \times \vec{P}}{\epsilon_0 [\epsilon(\omega) - 1]} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad (7)$$

Take the plane waves

$$P_y = P_{y0} e^{i(qx - \omega t)}, \quad H_z = H_{z0} e^{i(qx - \omega t)}$$

Step 1 — the $\vec{P}$ - $\vec{H}$ relation

$$\begin{aligned} \therefore \nabla \times \vec{P} &= -\epsilon_0 \mu_0 (\epsilon(\omega) - 1) \frac{\partial \vec{H}}{\partial t} \\ \Rightarrow \frac{\partial P_y}{\partial x} &= -\mu_0 \epsilon_0 (\epsilon(\omega) - 1) \dot{H}_z \\ &= +\mu_0 \epsilon_0 (\epsilon(\omega) - 1) H_{z0} e^{i(qx - \omega t)} (i\omega) \\ \Rightarrow P_{y0} e^{i(qx - \omega t)} (iq) &= \mu_0 \epsilon_0 (\epsilon(\omega) - 1) (i\omega) H_{z0} e^{i(qx - \omega t)} \end{aligned}$$

$$q P_{y0} - \omega \mu_0 \epsilon_0 [\epsilon(\omega) - 1] H_{z0} = 0 \quad (8)$$

Step 2 — the $\vec{H}$ - $\vec{P}$ relation

$$\vec{D} = \epsilon_0 \epsilon \vec{E} = \epsilon_0 \epsilon \frac{\vec{P}}{(\epsilon - 1)\epsilon_0}$$

$$\begin{aligned} \underbrace{\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}}_0 &= \frac{\partial}{\partial t} \left\{ \frac{\epsilon(\omega)}{\epsilon(\omega) - 1} \right\} P_y \\ \Rightarrow -H_{z0} e^{i(qx - \omega t)} (iq) &= \frac{\epsilon(\omega)}{[\epsilon(\omega) - 1]} P_{y0} e^{i(qx - \omega t)} (-i\omega) \end{aligned}$$

$$q H_{z0} - \frac{\omega \epsilon(\omega)}{[\epsilon(\omega) - 1]} P_{y0} = 0 \quad (9)$$

Step 3 — solve the pair Equations (8) and (9) written together:

$$\begin{aligned} q P_{y0} - \omega \mu_0 \epsilon_0 (\epsilon(\omega) - 1) H_{z0} &= 0 \\ -\frac{\omega \epsilon(\omega)}{[\epsilon(\omega) - 1]} P_{y0} + q H_{z0} &= 0 \end{aligned}$$

A non-trivial solution requires the determinant to vanish:

$$\begin{aligned} \begin{vmatrix} q & -\omega \mu_0 \epsilon_0 (\epsilon(\omega) - 1) \\ -\frac{\omega \epsilon(\omega)}{(\epsilon(\omega) - 1)} & q \end{vmatrix} &= 0 \\ \Rightarrow q^2 - \frac{\omega^2 \mu_0 \epsilon_0 (\epsilon(\omega) - 1) \epsilon(\omega)}{(\epsilon(\omega) - 1)} &= 0 \quad \mu_0 \epsilon_0 = 1/c^2 \\ \Rightarrow q^2 = \omega^2 \epsilon(\omega) \mu_0 \epsilon_0 &= \frac{\omega^2 \epsilon(\omega)}{c^2} \end{aligned}$$

Polariton dispersion

$$\omega^2 = \frac{q^2 c^2}{\epsilon(\omega)} \quad (10)$$

Eq. (10) represents **coupled electromagnetic and mechanical waves**, called polaritons.

4 Ionic lattice insert $\varepsilon(\omega)$ into the polariton relation

Use the dispersion relation of an ionic lattice:

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{(\varepsilon_{st} - \varepsilon_\infty) \omega_0^2}{\omega_0^2 - \omega^2} \quad (11)$$

Use (11) in (10):

$$\omega^2 = \frac{q^2 c^2}{\varepsilon_\infty + \frac{(\varepsilon_{st} - \varepsilon_\infty) \omega_0^2}{(\omega_0^2 - \omega^2)}} \quad (12)$$

Combine the denominator:

$$\omega^2 = \frac{q^2 c^2 (\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2) \varepsilon_\infty + \omega_0^2 (\varepsilon_{st} - \varepsilon_\infty)} \quad \boxed{\text{target: } ax^2 + bx + c = 0}$$

$$\Rightarrow \omega_0^2 \omega^2 \varepsilon_\infty - \omega^4 \varepsilon_\infty + \omega^2 \omega_0^2 (\varepsilon_{st} - \varepsilon_\infty) = q^2 c^2 \omega_0^2 - q^2 c^2 \omega^2$$

$$\Rightarrow \varepsilon_\infty (\omega^2)^2 - \omega^2 (q^2 c^2 + \varepsilon_{st} \omega_0^2) + q^2 c^2 \omega_0^2 = 0$$

Solve the quadratic in ω^2

$$\begin{aligned} \omega^2 &= \frac{(q^2 c^2 + \varepsilon_{st} \omega_0^2) \pm \sqrt{(q^2 c^2 + \varepsilon_{st} \omega_0^2)^2 - 4 \varepsilon_\infty q^2 c^2 \omega_0^2}}{2 \varepsilon_\infty} \\ &= \frac{1}{2} \left[\frac{\varepsilon_{st}}{\varepsilon_\infty} \omega_0^2 + \frac{q^2 c^2}{\varepsilon_\infty} \right] \pm \frac{1}{2} \sqrt{\left(\frac{\varepsilon_{st}}{\varepsilon_\infty} \omega_0^2 + \frac{q^2 c^2}{\varepsilon_\infty} \right)^2 - \frac{4 \omega_0^2 q^2 c^2}{\varepsilon_\infty}} \end{aligned}$$

The dispersion we obtain

$$\omega^2 = \frac{1}{2} \underbrace{\left[\frac{\varepsilon_{st}}{\varepsilon_\infty} \omega_0^2 + \frac{q^2 c^2}{\varepsilon_\infty} \right]}_{\text{term (I)}} \pm \frac{1}{2} \underbrace{\left[\left(\frac{\varepsilon_{st}}{\varepsilon_\infty} \omega_0^2 + \frac{q^2 c^2}{\varepsilon_\infty} \right)^2 - \frac{4 \omega_0^2 q^2 c^2}{\varepsilon_\infty} \right]^{1/2}}_{\text{term (II)}}$$

5 Large q upper and lower branch

For large q , expand each term of the dispersion:

$$\text{term (I)} \simeq \frac{1}{2} \frac{q^2 c^2}{\varepsilon_\infty}$$

$$\text{term (II)} \simeq \frac{1}{2} \left[\frac{q^4 c^4}{\varepsilon_\infty^2} - \frac{4 \omega_0^2 q^2 c^2}{\varepsilon_\infty} \right]^{1/2}$$

$$= \frac{1}{2} \frac{q^2 c^2}{\varepsilon_\infty} \left[1 - \frac{4 \omega_0^2 \varepsilon_\infty}{q^2 c^2} \right]^{1/2}$$

$\leftarrow$ expand: the bracket term is large for $q \rightarrow \infty$

$$= \frac{1}{2} \frac{q^2 c^2}{\varepsilon_\infty} \left[1 - \frac{2 \omega_0^2 \varepsilon_\infty}{q^2 c^2} \right]$$

$$= \frac{1}{2} \frac{q^2 c^2}{\varepsilon_\infty} - \frac{q^2 c^2}{2 \varepsilon_\infty} \cdot \frac{2 \omega_0^2 \varepsilon_\infty}{q^2 c^2}$$

$$= \frac{1}{2} \frac{q^2 c^2}{\varepsilon_\infty} - \omega_0^2$$

Upper branch

$$\begin{aligned}\omega_+^2 &= \frac{1}{2} \frac{q^2 c^2}{\epsilon_\infty} + \frac{1}{2} \frac{q^2 c^2}{\epsilon_\infty} - \omega_0^2 \\ &= \frac{q^2 c^2}{\epsilon_\infty} - \omega_0^2 \sim \frac{q^2 c^2}{\epsilon_\infty}\end{aligned}$$

$$\therefore \boxed{\omega_+^2 = \frac{q^2 c^2}{\epsilon_\infty}} \text{ (photon-like)}$$

Lower branch

$$\begin{aligned}\omega_-^2 &= \cancel{\frac{1}{2} \frac{q^2 c^2}{\epsilon_\infty}} - \cancel{\frac{1}{2} \frac{q^2 c^2}{\epsilon_\infty}} + \omega_0^2 \\ &= \omega_0^2\end{aligned}$$

$$\boxed{\omega_-^2 = \omega_0^2} \text{ (phonon-like)}$$

Check and refinement (exact relations)

The quadratic $\epsilon_\infty(\omega^2)^2 - (q^2 c^2 + \epsilon_{st} \omega_0^2) \omega^2 + q^2 c^2 \omega_0^2 = 0$ has two roots whose sum and product are known exactly (Vieta):

$$\omega_+^2 + \omega_-^2 = \frac{q^2 c^2 + \epsilon_{st} \omega_0^2}{\epsilon_\infty}, \quad \omega_+^2 \omega_-^2 = \frac{q^2 c^2 \omega_0^2}{\epsilon_\infty}.$$

For large q these give $\omega_-^2 \rightarrow \omega_0^2$ and $\omega_+^2 = \frac{q^2 c^2}{\epsilon_\infty} + (\omega_L^2 - \omega_T^2) + \dots$. The note above dropped the constant $\epsilon_{st} \omega_0^2 / \epsilon_\infty$ inside term (I), which only shifts the constant: the leading photon-like behaviour $\omega_+^2 \simeq q^2 c^2 / \epsilon_\infty$ is unchanged. *Numerical check* ($\epsilon_{st} = 5$, $\epsilon_\infty = 2$, $\omega_0 = c = 1$, $q = 10$): exact $\omega_+^2 = 51.53$; the formula above gives $50 + 1.5 = 51.5$, while the bare $q^2 c^2 / \epsilon_\infty = 50$ is the leading term only.

6 Small q lower and upper branch

Start again from the full dispersion and let q be small, so $q^2 c^2 \ll \epsilon_{st} \omega_0^2$ inside the root:

$$\omega^2 \simeq \frac{1}{2} \frac{\epsilon_{st}}{\epsilon_\infty} \omega_0^2 \pm \frac{1}{2} \left[\left(\frac{\epsilon_{st} \omega_0^2}{\epsilon_\infty} \right)^2 - \frac{4 \omega_0^2 q^2 c^2}{\epsilon_\infty} \right]^{1/2}$$

Lower branch

$$\begin{aligned}\omega^2 &= \frac{1}{2} \frac{\epsilon_{st}}{\epsilon_\infty} \omega_0^2 - \frac{1}{2} \left[\frac{\epsilon_{st} \omega_0^2}{\epsilon_\infty} \left(1 - \frac{4 \omega_0^2 q^2 c^2 \epsilon_\infty}{\epsilon_\infty \epsilon_{st}^2 \omega_0^4} \right)^{1/2} \right] \\ &= \frac{1}{2} \frac{\epsilon_{st}}{\epsilon_\infty} \omega_0^2 - \frac{1}{2} \left[\frac{\epsilon_{st} \omega_0^2}{\epsilon_\infty} \left(1 - \frac{2 q^2 c^2 \epsilon_\infty}{\omega_0^2 \epsilon_{st}^2} \right) \right] \\ &= \cancel{\frac{1}{2} \frac{\epsilon_{st}}{\epsilon_\infty} \omega_0^2} - \cancel{\frac{1}{2} \frac{\epsilon_{st}}{\epsilon_\infty} \omega_0^2} + \frac{q^2 c^2 \epsilon_\infty}{\omega_0^2 \epsilon_{st}^2} \cdot \frac{\epsilon_{st} \omega_0^2}{\epsilon_\infty} \\ &= \frac{q^2 c^2}{\epsilon_{st}}\end{aligned}$$

Lower branch, small q

$$\omega^2 = \frac{q^2 c^2}{\epsilon_{st}} \quad \text{so } \omega \rightarrow 0 \text{ as } q \rightarrow 0 \text{ (photon-like)}$$

Upper branch As $q \rightarrow 0$ the square root is $\sqrt{\Delta} \simeq S$, so

$$\begin{aligned}\omega^2 &= \frac{1}{2} \left[\frac{\epsilon_{st}}{\epsilon_\infty} \omega_0^2 + \frac{q^2 c^2}{\epsilon_\infty} \right] + \frac{1}{2} \left[\frac{\epsilon_{st}}{\epsilon_\infty} \omega_0^2 + \frac{q^2 c^2}{\epsilon_\infty} \right] \\ &= \frac{\epsilon_{st}}{\epsilon_\infty} \omega_0^2 + \frac{q^2 c^2}{\epsilon_\infty} = \omega_L^2 + \frac{q^2 c^2}{\epsilon_\infty} \quad \text{(leading order)}\end{aligned}$$

Upper branch, small q

$$\omega \rightarrow \omega_L \text{ as } q \rightarrow 0 \quad \text{with} \quad \sqrt{\frac{\epsilon_{st}}{\epsilon_\infty}} \omega_0 = \omega_L$$

Refinement of the q^2 term

Using Vieta, $\omega_+^2 = S - \omega_-^2 = \omega_L^2 + \frac{q^2 c^2}{\epsilon_\infty} - \frac{q^2 c^2}{\epsilon_{st}} + \dots$, i.e.

$$\omega_+^2 \simeq \omega_L^2 + q^2 c^2 \left(\frac{1}{\epsilon_\infty} - \frac{1}{\epsilon_{st}} \right).$$

The statement $\omega_+ \rightarrow \omega_L$ stands; only the coefficient of q^2 needs the extra $-1/\epsilon_{st}$. (Check, $q = 0.3$: exact 2.5272; refined 2.5270; without the correction 2.545.)

7 Summary dispersion plot

	small q	large q
upper branch ω_+	$\sqrt{\omega_L^2 + q^2 c^2 (1/\epsilon_\infty - 1/\epsilon_{st})} \rightarrow \omega_L$	$qc/\sqrt{\epsilon_\infty}$ (photon-like)
lower branch ω_-	$qc/\sqrt{\epsilon_{st}}$ (photon-like)	$\omega_0 = \omega_T$ (phonon-like)

